

IDP PRACTICAL EXAMINATION

Duration: 60 minutes

Marks: 20

Create a Dart program with the following requirements.

No	Decription				Marks
R1	Create an abstract class named Stationery has an abstract method named exportData() returns the void type.				1
R2	Create a Calculator class that extends Stationery with the following requirements:				4
	No	Field Name	Type	Constraint	
	1	code	integer	Required	
	2	model	String	Required	
	3	price	double	Required, greater than zero	
4	qty	integer	Required, greater than zero		
R3	Create a class ShopStore has the following members: + Generic Map<Calculator> list. + Method addNew(Calculator calculator): - Creates an instance of Calculator class, - Input the code, model, price and qty (<i>using loop and the try on statement to handle until the values valid</i>), - Adds this instance into the generic list. + Method display(): <i>Display all calculators in list.</i> + Method delete(int code): <i>delete calculator by code. delete calculator by code. An error message will be raised if code is not found.</i> + Method saveToFile(): <i>to write Calculator list into a text file named "Calculators.csv".</i> - Note: <i>Information of each Calculator shoud be write in a new line.</i>				10
R4	Create a Dart file ShopApp that has the main method to test. + Use switch case and loop statements to shows a following menu and exits the application if only exit fuction is choosen 1. Add new: On typing "add" that invokes the addNew method, 2. Delete Calculator: On typing "delete" that invokes the delete method, 3. Display all: On typing "show" that invokes the display method, 4. Save: On typing "save" that invokes the saveToFile method, 5. Exit: On typing "exit" that the application exits safely				5

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Note: - All the marks are evaluated in the main method.
- AI tools are NOT ALLOWED during the exam.